

Uniform Deflection of Induction Vectors at the South Chilean Continental Margin: A Hint at Electrical Anisotropy in the Crust

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Abstract

Induction vectors at the South Chilean continental margin display a unique pattern throughout the study area between latitudes 38-41°S and longitudes 71-74°W: At long periods of approx. 3.000 s their real parts are uniformly deflected from the W-E direction (which would be expected due to the coast effect and/or anomalies beneath the roughly N-S striking Andean mountain chain) to the NE. Attempts to model this behavior with simple and geologically realistic 3-D models failed, but a reasonable data fit was obtained by employing 2-D models with an anisotropic upper crust. This anisotropy hints at a deeply fractured crust with strike directions running obliquely to the continental margin in accordance with a recently mapped pattern of fault zones in the arc and forearc.

Introduction

An amphibious long-period magnetotelluric experiment was accomplished in austral summer 2004/2005 in South-Central Chile (TIPTEQ project, see Kapinos et al., this volume) which complemented an earlier study carried out slightly further to the south (Brasse and Soyer 2001). Transfer functions in the period range between 10 s – 20.000 s from approx. 70 stations, set up between the Argentinian border and the Pacific Ocean (plus an extension across the Peru-Chile trench with seafloor sites) are available by now. On land, the network encompasses the areas of the Coastal Cordillera, the Central Depression or Longitudinal Valley and the Principal Cordillera, which constitutes the active volcanic arc (see Fig. 1).

Given the average N10°E trend of the trench, which reaches a maximum depth of ~4.600 m in the study area, and the similar overall course of the volcanic chain, it was expected that real parts of induction vectors would show a general W-E tendency. This is, however, not the case. Instead, at long periods, all real induction vectors point systematically NE for all sites in the measuring area, regardless on which geological unit data were collected (Fig. 1). Note that there is not a single site in the study area where this observation is opposed. On the other hand, this effect is not observed at shorter periods (Kapatos et al., this volume).

3-D Modeling to explain induction vectors

It is obvious that this behavior cannot be explained by pure 2-D models. We therefore tested several simple 3-D approaches to account for the deflection over a large area, in particular the N-S extent of at least 350 km (it is not known if transfer functions continue to be this anomalous to the north and south of the measuring area, but this may quite safely be assumed). Such models must incorporate the Pacific Ocean with an average depth of ~4 km and an almost N-S running coastline, and some other structure of large, regional extent which accounts for the

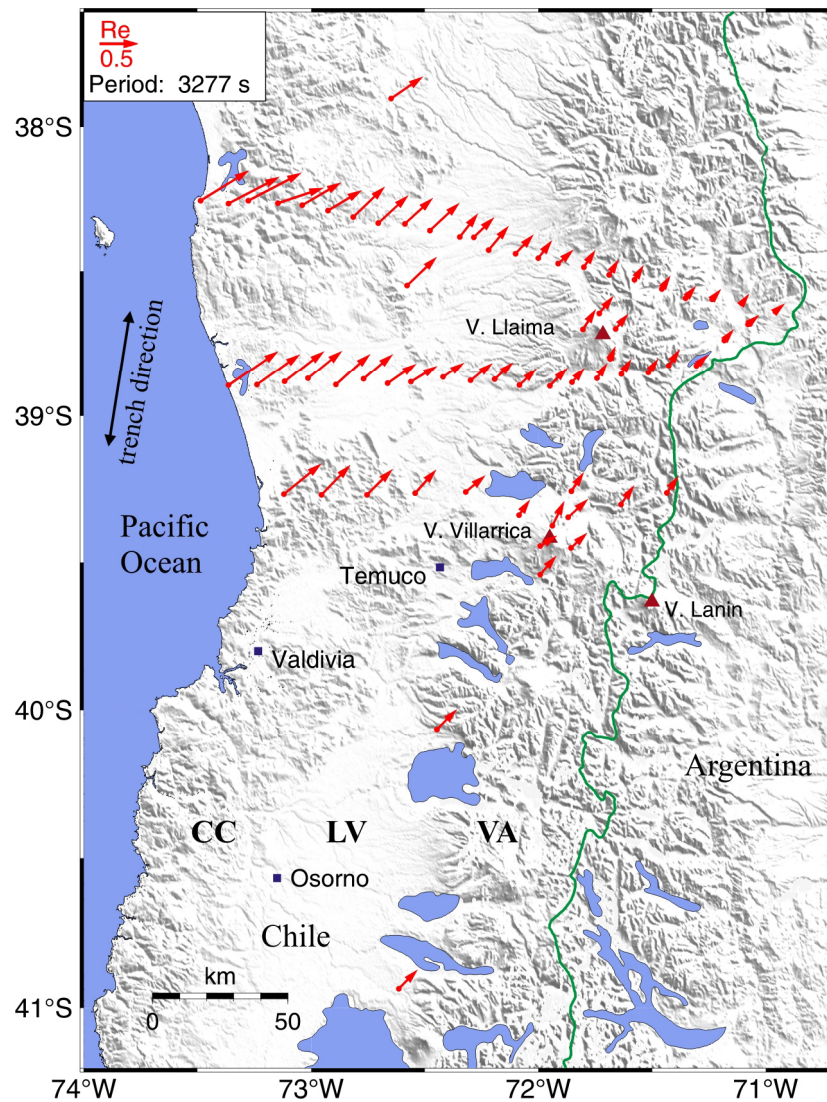


Fig. 1: Deflection of real induction vectors at a period of 3.277 s from the expected W-E direction in South Chile. CC Coastal Cordillera, LV Longitudinal Valley, VA volcanic arc. Topographic image is derived from SRTM-90 m data (NASA).

deflection. For the computations the algorithm of Mackie et al. (1994) was applied, seawater resistivity was fixed at $\rho = 0.3 \Omega\text{m}$.

Test 1: It may be possible that a layer with increasing conductance from south to north exists at some depth in the crust or even in the upper mantle. This may indeed explain the induction vectors but leads to inherently large conductancies in the northern part (and unrealistic conductivities if layer thickness is not changed accordingly). Furthermore there exists no geological evidence whatsoever for such a layer and the idea is thus abandoned here.

Test 2: South of the southernmost site begins archipelagic Chile, i.e., the Central Depression is submerged and the Coastal Cordillera becomes a chain of islands, with Isla Chiloé being the largest (not shown in Fig. 1). The distribution of seawater masses causes therefore a deflection of induction vectors near latitude 41°S, but this effect does not reach far enough to the north, taking the known water depths into account.

Test 3: At 45-46°S, i.e., 450-550 km south of the study area the Chile Rise is subducted beneath the continent; this is the triple junction between Antarctic, Nazca and South Ameri-

can plates. It may be speculated that the location of the triple junction is associated with a deep seated plume structure of enhanced conductivity. Such a hypothetical good conductor (resistivity $1 \Omega\text{m}$) was incorporated into the 3-D model (which also takes the irregular coastline into account); the response is shown in Fig. 2. Again, the model response is only compatible with the data at the southernmost sites of the measuring area; in the north vectors point strictly W-E.

Summarizing, simple and (geologically) realistic 3-D models explaining the observed induction vectors could not be found (which does of course not exclude more detailed classes of models, see later). We therefore tested another approach, the simulation of a deeply fractured crust with anisotropic 2-D models.

2-D models with anisotropy

It is an almost trivial exercise to construct a 2-D model with an anisotropic layer in the continental crust which may indeed explain the observed transfer function qualitatively. Such a simple model is displayed in Fig. 3, which incorporates the Pacific Ocean with a crude bathymetry, a homogeneous crust and upper mantle with a resistivity of $300 \Omega\text{m}$, and an anisotropic layer in the upper crust. Resistivity in x-direction is set to $\rho_x = 1 \Omega\text{m}$, and in y-direction to $\rho_y = 300 \Omega\text{m}$, while the vertical resistivity is again $\rho_z = 1 \Omega\text{m}$ (but note that this quantity is not resolved in MT). The ocean primarily accounts for the length of induction vectors near the coast while the strike of anisotropy is responsible for the deflection from W-E. In the model of Fig. 3 this anisotropy strike is set to -57° , i.e., the conductive axis is basically running NW-SE. This strike direction is also motivated by a recent mapping of fault structures in the forearc and arc regions of southern Chile (Melnick et al. 2006), which is discussed later in this contribution. Transfer functions were calculated employing the algorithm of Li (2002). Full anisotropy has to account for 6 variables, the principal resistivities, strike, dip and slant (cf. Pek and Verner 1997). Due to the resulting complexity we varied only the first 4 parameters leaving dip and slant constant at 90° and 0° , respectively. Since ρ_z has only minor influence this quantity was also set constant and equal to ρ_x for most model experiments. We restrict analysis to the landward stations along the central profile (Fig. 1) here.

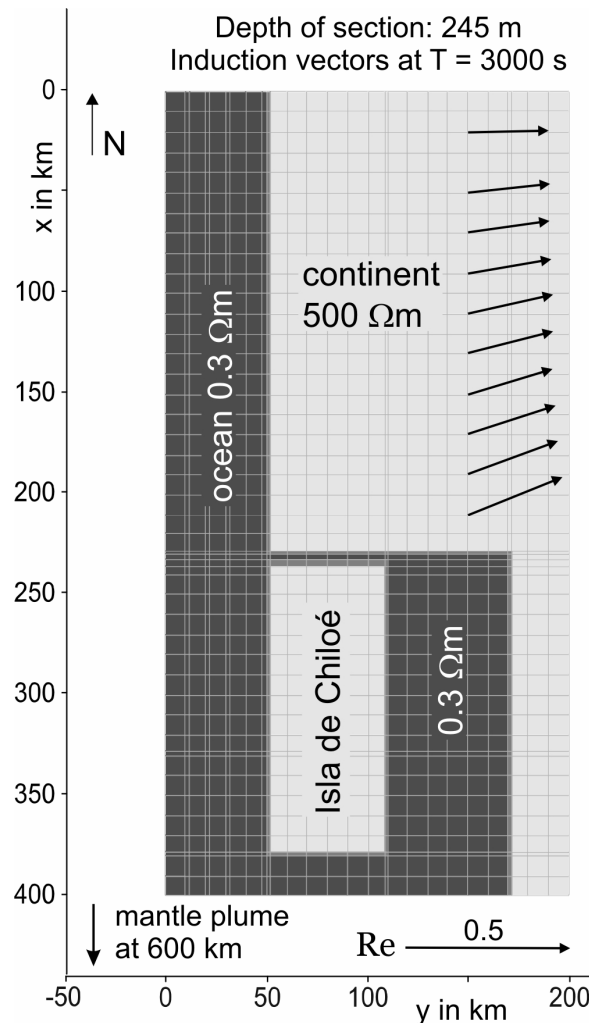


Fig. 2: Plan view of 3-D test model 3 at a depth of 245 m. Real induction vectors at a period of 3.000 s are simulated for a N-S profile in the Longitudinal Valley. The model comprises the ocean, the island of Chiloé, the Gulf of Ancud, and a deep mantle anomaly in the area of the triple junction 200 km to the south of the plot margin.

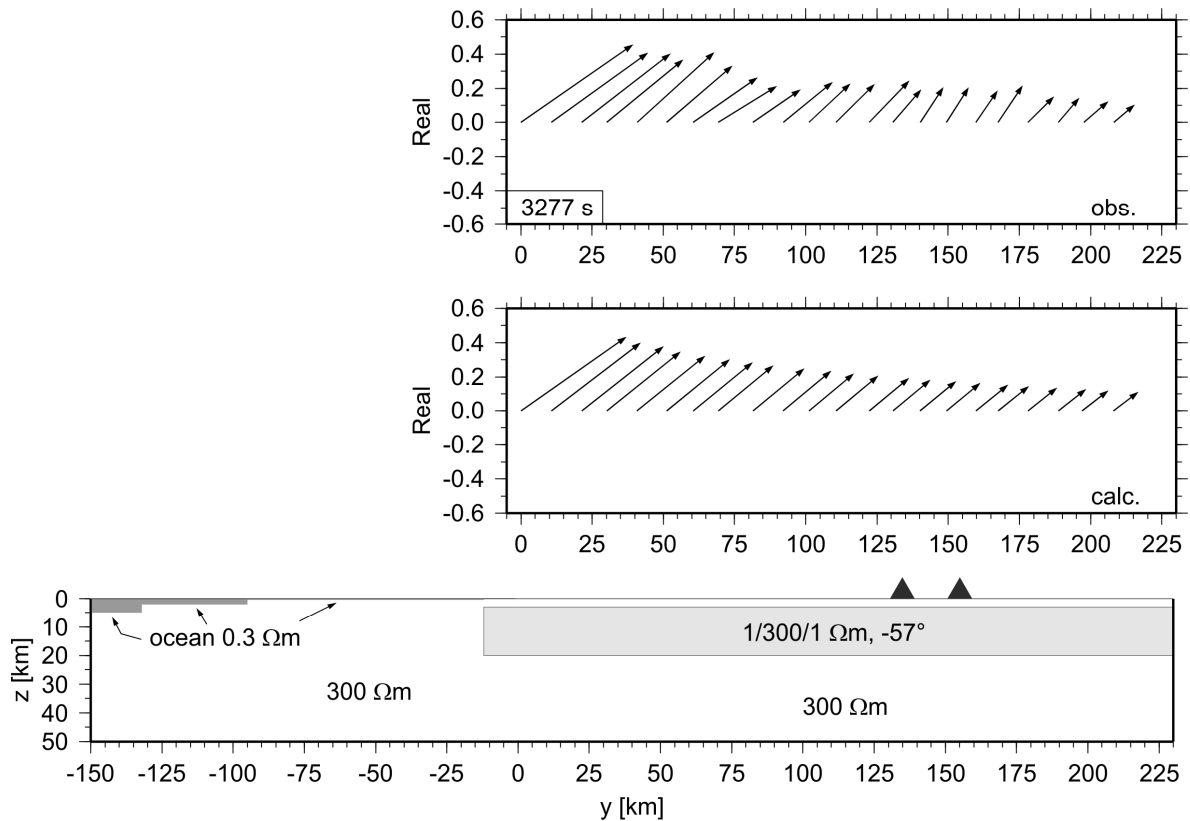


Fig. 3: Simple 2-D model incorporating the Pacific Ocean and an anisotropic layer in the upper crust (bottom). Numbers give resistivities in principal directions and anisotropy strike. Top: Observed real induction vectors along the middle profile in the study area, center: model response. 0 km corresponds to the coastline.

This simple model does already explain the observations with the exception of a minimum of vector length at 75 km (and some data scatter). We carried out numerous tests to constrain upper and lower boundaries of the anomalous layer. Without going into further detail here it may be concluded that one cannot resolve (as might be expected) the thickness of the anisotropic layer; it may well reach into the lower crust until Moho depths. The crust-mantle boundary lies at 35-45 km depth in South Chile as inferred from seismological studies (e.g., Krawczyk et al. 2006). On the other hand the upper crust has to show an anisotropic signature; anisotropy confined to the lower crust or even the upper mantle alone is not compatible with observations. It is not easy to constrain the upper limit of the layer since, at short periods, induction vectors show more local (also 3-D) features which are difficult to implement.

The minimum of vector lengths at ~ 75 km may be accounted for by introducing a homogeneous and isotropic block, as shown in Fig. 4a. So far this model yields the best fitting response. Note that anisotropy persists along the whole profile and even extends below the volcanic arc (where induction vectors are small but still deflected) and into Argentina, east of the easternmost site location. The model in Fig. 4a also includes well-conducting sediments ($\rho = 10 \Omega\text{m}$) in the Central Depression and the Bío-Bío valley. The thickness of these sediments is not known and it was arbitrarily set to 2 km here. Finally, Fig. 4b shows the modeled and measured transfer functions for real and imaginary parts at all periods along the central profile. Besides responses at short periods (and at long periods for the imaginary part) the data fit seems reasonable.

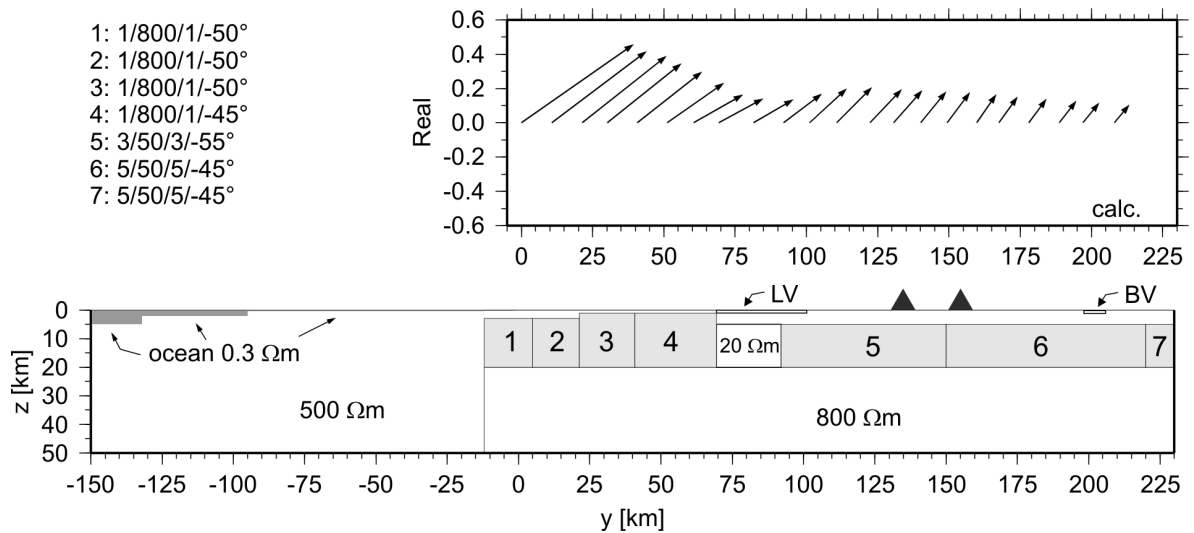


Fig. 4a: 2-D anisotropic model yielding the (so far) best fitting response. The anisotropic layer is broken by an isotropic block; in addition the sediments of the Longitudinal Valley (LV) and below the Bío-Bío river (BV) are introduced as a good conductor. Upper left corner shows resistivities of anisotropic blocks in principal directions and strike angle.

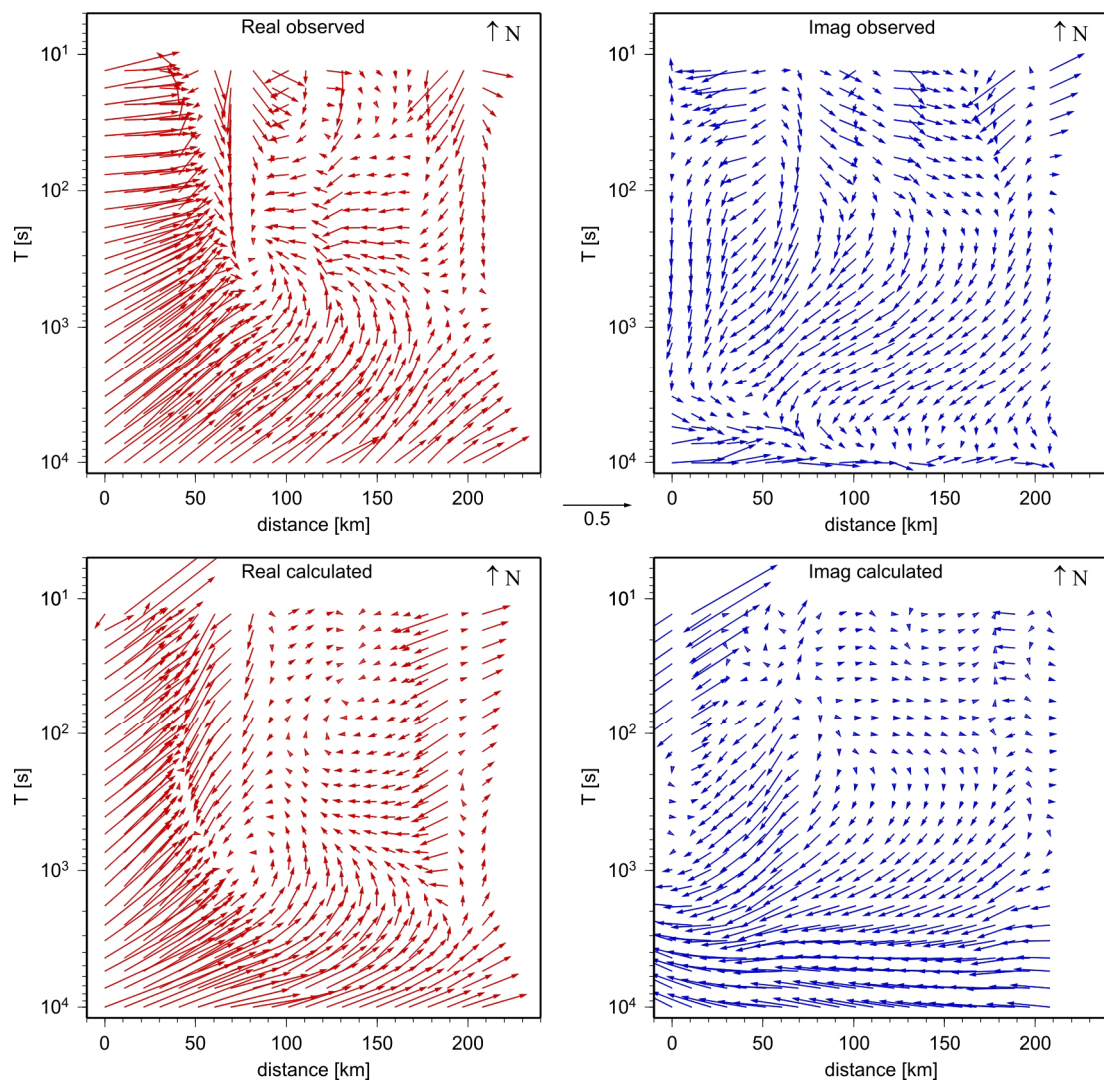


Fig. 4b: Comparison of model response (bottom) with data (top) for all periods along the profile (left: real, right: imag. parts).

Discussion

The overall preference direction of electrical conductivity agrees well with the pattern of faults in the South-Central Andes, striking obliquely to the continental margin as shown in Fig. 5 (see Melnick et al. 2006). The most prominent one is the N10°E striking Liquiñe-Ofqui Fault Zone (LOFZ) which extends from the triple junction area until 38°S, where it terminates at the Bío-Bío-Aluminé Fault (BBAF). The LOFZ is assumed to largely control volcanism in S. Chile (e.g., Cembrano et al. 1996) because many of the Quaternary and active volcanoes are aligned along this lineament. But electrically this mega shear zone is apparently of much less importance than the NW-SE striking fault system, which might be regarded as offering pathways for deep fluids. The most prominent ones crossing the MT profiles are mapped, e.g., as Lanalhue Fault (LFZ) and Mocha-Villarrica Fault (MVZ). The anisotropic model hints at an even more fractured upper (perhaps also lower) crust; individual faults are not resolvable with this approach. The fact that all stations in S. Chile display induction vectors pointing obliquely to the trench and the volcanic front will make it difficult to resolve single faults with full, isotropic 3-D models. Further stations in the surroundings of promising candidates

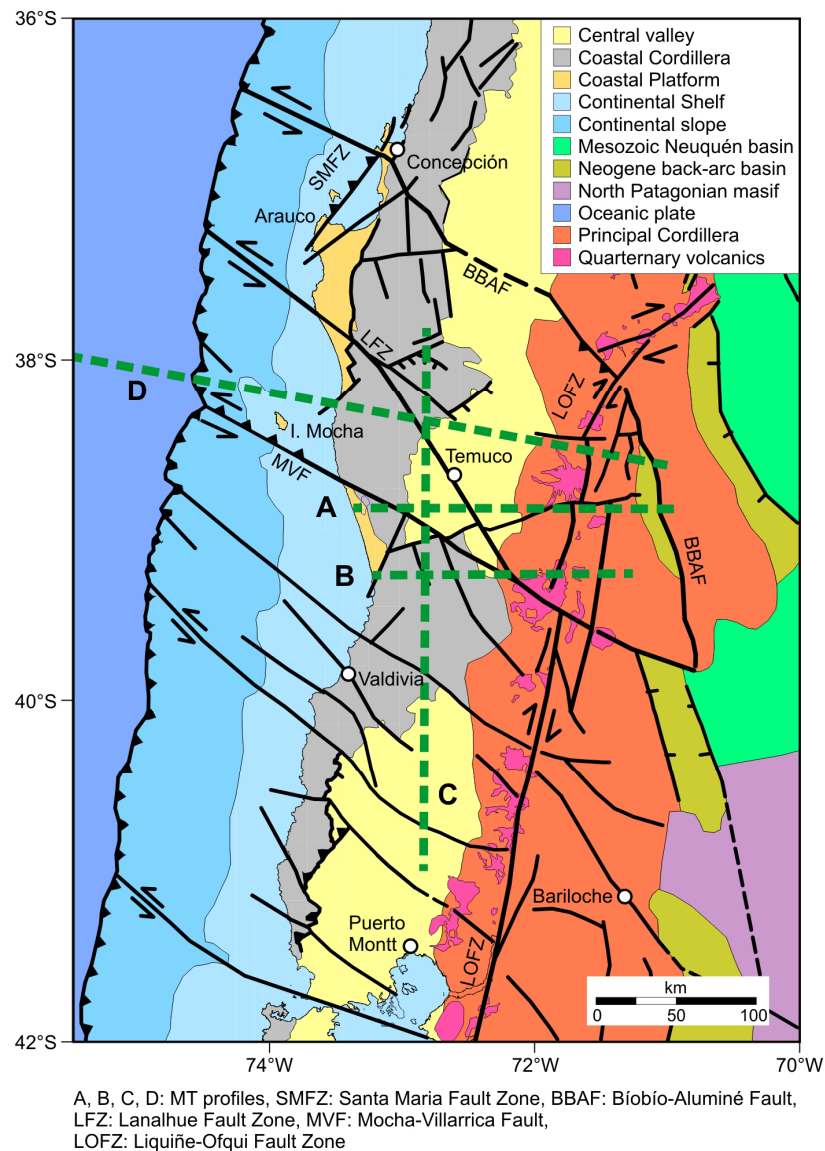


Fig. 5: Geological and tectonic map of the South-Central Andes, modified after Melnick et al. (2006). A-D are the MT profiles measured in 2000 and 2004/05.

(e.g., Lanalhue Fault) would be necessary to accomplish this task.

Unfortunately the model in Fig. 4 explains the impedances only in a crude manner. For the time being we have been unable to construct a model which satisfactorily fits both magnetotelluric and vertical magnetic field observations; this constitutes the next task for the evaluation of the data set. However, several features of the model presented here correspond to the isotropic impedance model by Brasse and Soyer (2001), in particular the enhanced conductivity below the Central Depression and the generally higher conductivities beneath the volcanic arc.

The characteristics of induction vectors outside the area depicted in Figs. 1 and 5 is not known and it would thus be a rewarding effort to establish further sites to the north of the Bío-Bío Fault and to the south in archipelagic Chile.

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